

# TrustBallot — Product Requirements Document (PRD)

## Version 1.2 | 3-Month Delivery Cycle

Revision note: v1.2 closes the voter-auth ownership gap, fixes cross-document inconsistencies (owner of metadata obfuscation, language count), reframes over-claimed metrics, and adds newly acknowledged risks/limitations. See changelog at end.

## 1. Problem Statement

India has ~300M internal migrants unable to vote in home constituencies due to cost/distance of return travel. Existing ECI RVM prototype (2022) is non-networked and doesn't solve double-voting or verifiability. TrustBallot delivers a networked, cryptographically-secure kiosk voting system.

## 2. Goals & Success Metrics

| Goal                                   | Metric                                                                                                                                             |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Prevent double voting                  | 100% nullifier-check accuracy <b>within one IBC sync cycle (~2 hrs)</b> — <i>not</i> real-time across kiosk/EVM boundary (see Risks §8.6)          |
| Preserve ballot secrecy                | Zero vote-to-voter linkage in audit, under a <b>classical (non-quantum) adversary model</b> for the HE layer (see §8.7)                            |
| Kiosk usability for low-literacy users | <3 min average voting time, symbol-based nav tested with 20+ users, <b>including constituencies with 15-30+ candidates (scroll-tested)</b>         |
| System resilience                      | BFT consensus survives 1-node failure without halting, <b>demonstrated across physically/virtually separate node instances</b> (not one shared VM) |
| Demo-readiness                         | End-to-end simulated election (auth → ZKP → tally) functional by Month 3                                                                           |

## 3. Users & Personas

- Migrant Voter:** low/no technical literacy, regional language speaker, time-constrained.
- Polling/ECI Officer:** manual-override authority, kiosk on-site supervisor.
- Auditor/Judiciary/Civil-society Node Operator:** verifies consensus, runs BFT node.

## 4. Scope (3-Month MVP)

**In scope:** Kiosk auth simulation (QR + signature + stubbed face-match), ZKP-based vote proof, nullifier system, consortium blockchain (4-node, physically separated), homomorphic tally, basic UI (**2 languages, fixed**), Duress PIN, kiosk pubkey registration (Phase 0), voter-facing error/retry states.

**Out of scope (Future/Post-MVP):** Real EPIC/DigiLocker integration, real face-match ML model (stubbed for demo — see §5 note), real hardware attestation supply chain, legal/RoPA amendment work, nationwide EVM interoperability, production HSM key management, nullifier ledger pruning at scale, post-quantum homomorphic encryption.

## 5. Feature List & Priority

All cryptography consolidated under **Anushka**, per team decision. Auth is split: signature verification (crypto) vs. scan/UI/biometric-stub (kiosk client).

| Priority | Feature                                                                  | Owner                                     | Area                |
|----------|--------------------------------------------------------------------------|-------------------------------------------|---------------------|
| P0       | Kiosk pubkey registration (Phase 0, pre-election)                        | Anushka (crypto) + Shashwat (infra)       | Blockchain/Infra    |
| P0       | EPIC QR scan + camera UI                                                 | Arnav                                     | Frontend/Kiosk      |
| P0       | ECI public-key signature verification of EPIC/QR                         | <b>Anushka</b>                            | Blockchain (crypto) |
| P0       | Face-match — <b>stubbed pass/fail toggle for MVP demo</b>                | Arnav                                     | Frontend/Kiosk      |
| P0       | ZKP circuit (ownership, membership, one-hot, nullifier)                  | Anushka                                   | Blockchain          |
| P0       | Consortium chain + smart contract nullifier check                        | Anushka                                   | Blockchain          |
| P0       | Homomorphic tally + threshold decryption (t=3-of-4)                      | Anushka                                   | Blockchain          |
| P1       | Kiosk UI (regional lang ×2, symbols, Duress PIN)                         | Arnav                                     | Frontend            |
| P1       | Voter-facing error/retry flow on contract rejection                      | Arnav                                     | Frontend            |
| P1       | BFT consensus simulation (4 nodes, separate instances)                   | Shashwat                                  | Backend/Infra       |
| P1       | Dilithium post-quantum signing                                           | Anushka                                   | Blockchain          |
| P1       | Metadata obfuscation (batched shuffle, 50–100 votes, 2-hr timeout flush) | <b>Arnav</b>                              | Frontend/Middleware |
| P2       | EVM signaling / IBC sync simulation                                      | Shashwat                                  | Backend             |
| P2       | Disputed Vote flagging (procedural — no biometric retained)              | Shashwat (backend) + Arnav (dashboard UI) | Backend/Frontend    |
| P2       | Admin/audit dashboard                                                    | Arnav                                     | Frontend            |

**Note on face-match:** This is a computer-vision task, not cryptography, and has no assigned owner or stack entry beyond this line. It is explicitly stubbed (mock pass/fail) for the Month-3 demo. Building a real on-device face-match model is out of scope for this cycle.

## 6. Future Features (Post-3-Month Roadmap)

Real DigiLocker e-EPIC + ECI public key integration

Real face-match ML model (replacing MVP stub)

Hardware attestation & remote firmware verification pipeline

Multi-state pilot with live ECI electoral roll sync (dynamic Merkle tree updates — MVP uses a frozen snapshot)

Formal legal/policy toolkit for RoPA amendment advocacy

Full accessibility audit (screen readers, motor-impairment kiosk modes)

Mobile companion app for voter status tracking (non-voting, informational only)

Nullifier ledger pruning/archival strategy for national scale

Post-quantum homomorphic encryption research track

## 7. Team Roles

**Anushka — Blockchain & Crypto lead (all cryptography):** ZKP circuits, nullifier design, smart contract logic, homomorphic tallying, Dilithium signing, EPIC/QR signature verification, kiosk pubkey registration crypto, PLONK setup (see TRD §4 — avoids trusted-setup ceremony risk of Groth16).

**Shashwat — Backend/Infra lead:** BFT consensus simulation (on separate node instances), node networking, EVM/IBC sync logic, disputed-vote backend logic, kiosk registration infra, DevOps.

**Arnav — Frontend/UX & Middleware lead:** Kiosk UI (scan, PIN, candidate grid w/ pagination, error states), face-match stub, metadata obfuscation middleware (moved from Shashwat in v1.1, **now consistent across all docs**), Duress PIN flow, admin/audit dashboard.

**Cross-training mitigation (new in v1.2):** Given Anushka is sole owner of all cryptographic components (single point of failure — see Risks §8.4), Shashwat shadows ZKP circuit work during Month 1 Week 2–3 as a documented backup, without taking primary ownership.

## 8. Risks

ZKP proof generation latency on low-end kiosk hardware.

Scope creep from legal/policy dependencies outside technical control.

Team bandwidth given 3-month academic timeline.

**Anushka is a single point of failure** for all cryptographic components (ZKP, nullifier, contract, HE, Dilithium, signature verification). Mitigated by Month-1 cross-training shadow (§7), not eliminated. Cross-domain handoff risk — Arnav's middleware (metadata obfuscation) interacts closely with Shashwat's networking layer; requires clear interface contracts.

**Sync-window double-vote gap:** kiosk↔EVM nullifier collisions are only caught every ~2 hrs (IBC batch sync), not in real time. Success metric reframed accordingly (§2) rather than overclaiming real-time prevention.

**Ballot secrecy is not post-quantum secure** — Paillier HE can be harvested now and decrypted later by a future quantum computer, even though Dilithium signatures remain quantum-safe. Documented as an explicit MVP scoping limitation, not a full quantum-resistance claim.

**Verifiability vs. receipt-freeness tension (unsolved):** withholding ballot receipts (Design Spec) to prevent coercion inherently limits the "end-to-end verifiability" claim. This is an open problem in e-voting generally (unresolved in Estonia/Switzerland's systems too) — acknowledged, not solved, in this MVP.

**No revote mechanism after Duress PIN use (unsolved):** once a nullifier is spent on a spoiled ballot, the voter has no path to cast a real vote later. Structurally difficult to fix without reopening double-vote or coercion-linkage risk. Named as an open limitation.

**ECI centralization of gas/relay infra** creates a practical (non-cryptographic) chokepoint in an otherwise "trustless" consortium model. No fix within 3-month academic scope — named as a known centralization risk.

**Vote-buying via self-recorded screen capture:** Duress PIN protects against a coercer physically present, not a voter who records their own session to prove a vote remotely. Known unsolved weakness of remote voting generally.

Kiosk hardware supply-chain attestation is a documented dependency, not built in MVP (unchanged from v1.0/v1.1).

## Changelog (v1.1 → v1.2)

Assigned owner for EPIC QR/signature verification (Anushka) and scan/UI/biometric stub (Arnav) — previously unowned.

Fixed metadata obfuscation owner inconsistency across PRD/TRD/Charter (now Arnav everywhere).

Fixed language count inconsistency (locked to 2, Design Spec updated to match).

Reframed double-voting and ballot-secrecy metrics to match what the architecture actually guarantees.

Added kiosk pubkey registration (Phase 0), threshold decryption parameter ( $t=3$ -of-4), 2-hr batch timeout flush, voter-facing error/retry flow, candidate-grid pagination.

Added Risks §8.4–8.11 (SPOF, sync-window gap, PQ-HE gap, receipt-freeness tension, no-revote gap, centralization, vote-buying).